

Daily Progress Report — July 7, 2026

SMA + Stigmergy 3D Drone Swarm Simulation

Summary

Today's session was a comprehensive review and documentation session covering the full project lifecycle — from environment setup through algorithm implementation, convergence validation, and final report compilation. All five weekly internship reports were finalized, the research paper abstract, introduction, and conclusion were completed, and the complete project documentation was consolidated for submission.

Work Done Today

1. Weekly Report Finalization

Completed and formatted all five weekly internship reports covering:

- Week 1 — Environment setup, WSL2 configuration, algorithm study
- Week 2 — Networking resolution, MAVSDK setup, 2D validation
- Week 3 — Algorithm debugging, five bug fixes, convergence to fitness 1.780
- Week 4 — Metrics extraction, pheromone analysis, obstacle avoidance verification
- Week 5 — Report compilation, parameter review, codebase cleanup

2. Research Paper Sections Completed

- **Abstract** — finalized covering SMA+Stigmergy hybrid approach, jamming-resilient navigation, and SITL validation
- **Introduction** — completed covering motivation, communication dependency problem, biological stigmergy inspiration, proposed approach, and paper organization
- **Conclusion** — written covering key contributions, convergence results, simulation stack overview, and future work directions

3. Algorithm Documentation Review

Reviewed and confirmed accuracy of all algorithm documentation against the final working codebase, ensuring the mathematical formulation in the paper matches the implemented `sma_core.py`, `pheromone.py`, and `fitness.py` source files.

4. Project Q&A and Conceptual Clarification

Conducted a detailed review of all major project concepts including telemetry usage, NED coordinate system, offboard control flow, stigmergy mechanism, jamming zone behavior, and the three-behavior position update logic (random exploration, approach, contraction). This ensures all team members have a clear conceptual understanding before final submission.

Current Project Status

Component	Status
WSL2 + Gazebo + PX4 environment	Complete
Single-drone MAVSDK offboard control	Complete
2D algorithm validation	Complete
3D convergence (fitness 1.780)	Complete
Five sma_core.py bugs fixed	Complete
Metrics and pheromone analysis	Complete
Abstract and introduction	Complete
Conclusion	Complete
Weekly reports (all 5 weeks)	Complete
Drone 1 independent flight	Pending
Jamming comparison plot (paper evidence)	Pending

Remaining Tasks

- Resolve Drone 1 independent flight in Gazebo (MAVSDK port binding issue)
- Generate stigmergy-enabled vs disabled convergence comparison plots from logs/*.json
- Complete remaining paper sections (Related Work, Methodology, Results, Experimental Setup)
- Final codebase review and GitHub repository cleanup before submission

Files Referenced Today

File	Purpose
algorithm/sma_core.py	Core SMA position update logic
algorithm/pheromone.py	3D pheromone field implementation
algorithm/fitness.py	Fitness function and jamming zone detection
drone_control/connection.py	MAVSDK connect, arm, offboard helpers
drone_control/telemetry.py	NED position reader

File	Purpose
drone_control/setpoint.py	Position setpoint sender
config/settings.py	All algorithm hyperparameters
main.py	Entry point — full swarm control loop